

Software Requirements Specification

TaskFlow AI

Smart AI-Powered Project Management Platform

Version 1.0

Prepared for academic, portfolio, and hackathon submission

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for TaskFlow AI, a full-stack, AI-augmented project management platform. It is intended to guide design, development, and evaluation of the system across its four planned development phases, and to serve as a reference document for academic submission, portfolio presentation, or hackathon evaluation.

1.2 Scope

TaskFlow AI is a Trello-inspired collaborative project management system that combines Kanban-style task organization with AI-powered planning assistance and calendar synchronization. The system allows users to create workspaces, manage projects and tasks through a drag-and-drop board, collaborate with team members, and use an integrated AI assistant (powered by the Grok API) to break down projects, estimate timelines, prioritize work, and summarize progress. Task deadlines can be synchronized automatically with Google Calendar.

The system is built using React on the frontend and Spring Boot with PostgreSQL on the backend, secured with JWT-based authentication.

1.3 Intended Audience

- Development team implementing the platform
- Academic evaluators reviewing the project as a coursework or capstone submission
- Hackathon judges assessing technical scope and feasibility
- Future contributors or collaborators extending the system

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
JWT	JSON Web Token, used for stateless authentication
API	Application Programming Interface
ORM	Object-Relational Mapping
CRUD	Create, Read, Update, Delete
Kanban	Visual workflow management method using columns and cards
Grok API	Third-party AI/LLM service used for project planning assistance
JPA	Java Persistence API, used with Hibernate for ORM

1.5 References

- TaskFlow AI Project Blueprint (internal product specification document)
- Spring Boot, React, and PostgreSQL official documentation

2. Overall Description

2.1 Product Perspective

TaskFlow AI is a new, standalone, full-stack web application. It is not a modification of an existing system. The product follows a layered client-server architecture: a React single-page application communicates with a Spring Boot REST API, which persists data in a PostgreSQL database and integrates with two external services — the Grok AI API for intelligent planning features, and the Google Calendar API for deadline synchronization.

2.2 Product Functions

At a high level, the system shall allow users to:

- Register, authenticate, and manage their account securely
- Create and manage workspaces and invite team members
- Create projects and organize work using Kanban boards
- Create, assign, and track tasks with checklists, labels, and comments
- Use AI assistance to break down projects, write task descriptions, estimate deadlines, plan sprints, and assess project health
- Sync task deadlines automatically with Google Calendar
- View dashboards and generate reports on progress and productivity
- Receive notifications about task assignments, deadlines, and AI-generated plans

2.3 User Classes and Characteristics

User class	Description
Workspace owner / admin	Creates workspaces, manages members and roles, has full project control
Team member	Creates and manages tasks within assigned projects, collaborates with others
Individual user	Uses the platform solo for personal project and task tracking

2.4 Operating Environment

- Client: Modern web browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile
- Backend server: Java 17+ runtime with Spring Boot
- Database server: PostgreSQL 14+
- External dependencies: Grok API and Google Calendar API, accessed over HTTPS
- Calendar sync is dependent on Google Calendar API quotas and OAuth consent

3. System Features (Functional Requirements)

Functional requirements are grouped by module. Priority is rated as High (core, required for MVP), Medium (important but deferrable), or Low (optional enhancement).

3.1 Authentication Module

ID	Requirement	Priority
FR-1.1	The system shall allow a new user to register with name, email, and password.	High
FR-1.2	The system shall authenticate users via email and password and issue a JWT on success.	High
FR-1.3	The system shall allow users to log out, invalidating the client-side session token.	High

3.2 Dashboard Module

ID	Requirement	Priority
FR-2.1	The system shall display total projects, active projects, and tasks due today.	High
FR-2.2	The system shall display completed and pending task counts.	High
FR-2.3	The system shall display upcoming deadlines and a productivity percentage.	Medium
FR-2.4	The system shall display recent team activity on the dashboard.	Medium

3.3 Workspace Module

ID	Requirement	Priority
FR-3.1	The system shall allow a user to create, edit, and delete a workspace.	High
FR-3.2	The system shall allow a workspace owner to invite members by email.	High
FR-3.3	The system shall support role-based access control within a workspace (owner, admin, member).	High

3.4 Project Module

ID	Requirement	Priority
FR-4.1	The system shall allow users to create, edit, archive, and delete projects within a workspace.	High
FR-4.2	The system shall track project status, deadline, and progress percentage.	High
FR-4.3	The system shall allow users to mark a project as a favorite.	Low

3.6 Task Module

ID	Requirement	Priority
FR-6.1	The system shall allow creation of tasks with name, description, priority, assignee, and due date.	High
FR-6.2	The system shall support checklists within a task.	High
FR-6.3	The system shall support labels, attachments, and comments on a task.	Medium
FR-6.4	The system shall track estimated vs. actual hours and maintain an activity history per task.	Medium

3.7 Team Collaboration Module

ID	Requirement	Priority
FR-7.1	The system shall allow assigning one or more members to a task.	High
FR-7.2	The system shall support @mentions of members within comments.	Medium
FR-7.3	The system shall maintain an activity timeline per project and per task.	Medium

3.9 AI Integration Module (Grok API)

ID	Requirement	Priority
FR-9.1	The system shall generate a task breakdown from a single project description entered by the user.	High
FR-9.2	The system shall generate a detailed task description from a short task title.	High
FR-9.3	The system shall suggest a priority level for a given task.	Medium
FR-9.4	The system shall estimate a project completion timeline based on scope and team size.	Medium
FR-9.5	The system shall generate a multi-week sprint plan from a target deadline.	Medium
FR-9.6	The system shall generate a daily summary of completed tasks, upcoming deadlines, and at-risk items.	Low
FR-9.7	The system shall compute and display a project health score with contributing reasons.	Low

3.10 Google Calendar Integration Module

ID	Requirement	Priority
FR-10.1	The system shall automatically create a Google Calendar event when a task is given a due date.	Medium
FR-10.2	The system shall update the corresponding calendar event when a task's due date changes.	Medium
FR-10.3	The system shall delete the corresponding calendar event when a task is deleted.	Medium

ID	Requirement	Priority
FR-10.4	The system shall display upcoming calendar events within the dashboard.	Medium
FR-10.5	The system shall provide a one-click “Add to Calendar” action for meetings and deadlines.	Low

4. External Interface Requirements

4.1 User Interfaces

- A responsive React-based web interface covering authentication, dashboard, workspace, task detail, calendar, and reports views
- Charts rendered using Chart.js or Recharts

4.2 Hardware Interfaces

No specialized hardware is required. The system runs on standard client devices (desktop, laptop, tablet, mobile) with a modern web browser and internet connectivity.

4.3 Software Interfaces

Interface	Purpose
Grok API	Provides AI-generated task breakdowns, descriptions, priorities, and summaries
Google Calendar API	Synchronizes task deadlines and events with the user's calendar
PostgreSQL	Primary relational data store, accessed via Spring Data JPA

4.4 Communication Interfaces

- All client-server communication occurs over HTTPS using a RESTful JSON API
- Authentication tokens (JWT) are transmitted via Authorization headers
- Real-time collaboration features (Phase 4) shall use WebSockets

5. Non-Functional Requirements

5.1 Performance

- Standard API responses should return within 500ms under normal load
- AI-dependent endpoints (Grok-backed) should return within 5 seconds, with a loading state shown to the user

5.2 Security

- Passwords shall be stored using a strong one-way hash
- All endpoints except authentication shall require a valid JWT
- Role-based authorization shall be enforced at the API layer, not just the UI
- Sensitive API keys (Grok, Google) shall be stored as environment variables, never committed to source control

5.3 Reliability and Availability

- The system shall handle third-party API failures (Grok, Google Calendar) gracefully, without blocking core task management functionality
- Database transactions affecting multiple tables (e.g. task creation with calendar sync) shall be handled so partial failures do not corrupt data

5.4 Usability

- The interface shall be responsive across desktop and mobile viewports
- Core actions (creating a task, moving a card) shall require no more than two interactions

5.5 Maintainability and Scalability

- The backend shall follow a layered architecture (controller, service, repository) to support independent testing and extension
- The data model shall support horizontal growth in workspaces, projects, and tasks without schema redesign

8. Technology Stack

Layer	Technology
Frontend	React
Backend	Spring Boot
Database	PostgreSQL
Authentication	JWT + Spring Security
ORM	Spring Data JPA / Hibernate
Build tool	Maven
AI API	Grok API
Calendar API	Google Calendar API
Charts	Chart.js / Recharts
Drag and drop	@dnd-kit

9. Development Roadmap

Phase 1 — Foundation

- User authentication
- Workspace management
- Project CRUD
- Task CRUD
- PostgreSQL integration

Phase 2 — Collaboration

- Team members and roles
- Comments and labels
- Notifications
- Activity logs
- Dashboard

Phase 3 — Smart Features

- Grok API integration: project breakdown, task descriptions, priority suggestions, sprint planning, health score
- Google Calendar integration: deadline sync, calendar events, meeting scheduling

Phase 4 — Advanced Enhancements

- Analytics dashboard
- File attachments
- Export reports (PDF/CSV)
- Dark mode
- Optional weather API
- Real-time collaboration via WebSockets